

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 24

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Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.5, 16.6 (curl, divergence, parametric surfaces)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

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ALVARO: Start the recording!



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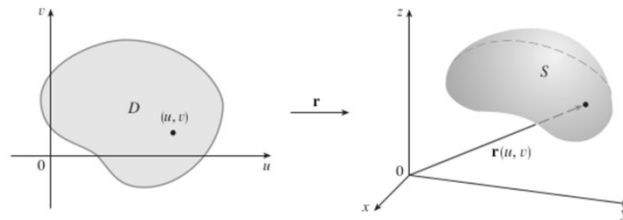
Surface Area

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About Parametrized Surfaces and their Area

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

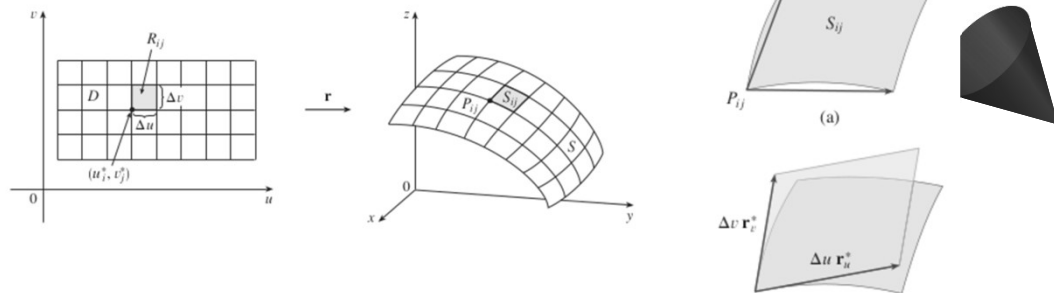
A parametric surface



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About Parametrized Surfaces and their Area

The image of the subrectangle R_{ij} is the patch S_{ij} .



$$|(\Delta u \mathbf{r}_u^*) \times (\Delta v \mathbf{r}_v^*)| = |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

$$\sum_{i=1}^m \sum_{j=1}^n |\mathbf{r}_u^* \times \mathbf{r}_v^*| \Delta u \Delta v$$

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About Parametrized Surfaces and their Area

If a smooth parametric surface S is given by the equation

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k} \quad (u, v) \in D$$

and S is covered just once as (u, v) ranges throughout the parameter domain D , then the **surface area** of S is

$$A(S) = \iint_D |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

where

$$\mathbf{r}_u = \frac{\partial x}{\partial u} \mathbf{i} + \frac{\partial y}{\partial u} \mathbf{j} + \frac{\partial z}{\partial u} \mathbf{k} \quad \mathbf{r}_v = \frac{\partial x}{\partial v} \mathbf{i} + \frac{\partial y}{\partial v} \mathbf{j} + \frac{\partial z}{\partial v} \mathbf{k}$$

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Surfaces Area of a Graph (and Length of a Graph)

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} \, dA$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

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Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$.



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Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$.

[Extra Space]



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Surfaces Area of a Surface of Revolution



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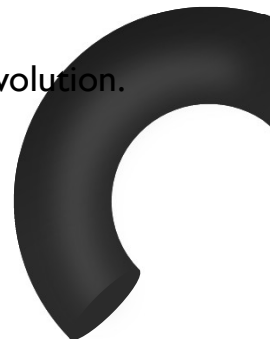
Surfaces Area of a Surface of Revolution

$$\begin{aligned}
 A &= \iint_D |\mathbf{r}_x \times \mathbf{r}_\theta| \, dA \\
 &= \int_0^{2\pi} \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx \, d\theta \\
 &= 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} \, dx
 \end{aligned}$$



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Example: Compute the surface area of the paraboloid
 $z = x^2 + y^2$ below the plane $z = 9$, as a surface of revolution.



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[Extra Space]

Example: Compute the surface area of the paraboloid
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Questions?

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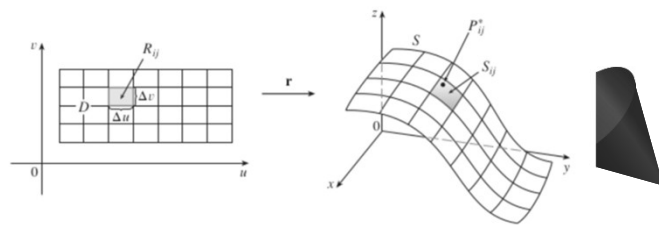
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Surface Integrals

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Integrals on Parametrized Surfaces

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$



$$\iint_S f(x, y, z) \, dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(P_{ij}^*) \, \Delta S_{ij}$$

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA$$

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Example: Compute the surface integral $\iint_S x^2 dS$ where S is the unit sphere.



$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$



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[Extra Space]

Example: Compute the surface integral $\iint_S x^2 dS$ where S is the unit sphere.



$$x = a \sin \phi \cos \theta \quad y = a \sin \phi \sin \theta \quad z = a \cos \phi$$



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[Extra Space]

Example: Compute the surface integral $\iint_S x^2 dS$ where S is the unit sphere.

$$\begin{aligned}
 \iint_S x^2 dS &= \iint_D (\sin \phi \cos \theta)^2 |\mathbf{r}_\phi \times \mathbf{r}_\theta| dA \\
 &= \int_0^{2\pi} \int_0^\pi \sin^2 \phi \cos^2 \theta \sin \phi d\phi d\theta = \int_0^{2\pi} \cos^2 \theta d\theta \int_0^\pi \sin^3 \phi d\phi \\
 &= \int_0^{2\pi} \frac{1}{2}(1 + \cos 2\theta) d\theta \int_0^\pi (\sin \phi - \sin \phi \cos^2 \phi) d\phi \\
 &= \frac{1}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{2\pi} \left[-\cos \phi + \frac{1}{3} \cos^3 \phi \right]_0^\pi = \frac{4\pi}{3}
 \end{aligned}$$

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Integrals on Graphs of Functions

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k}$$

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

$$\iint_S f(x, y, z) dS = \iint_D f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA$$

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Example: Compute the surface integral $\iint_S y \, dS$ where S is the surface
 $z = x + y^2$, $0 \leq x \leq 1$, $0 \leq y \leq 2$.

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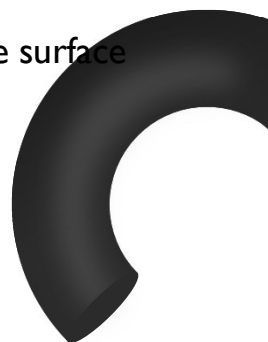
Example: Compute the surface integral $\iint_S y \, dS$ where S is the surface
 $z = x + y^2$, $0 \leq x \leq 1$, $0 \leq y \leq 2$.

[Extra Space]

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Example: Compute the surface integral $\iint_S z \, dS$ where S is the surface whose sides are given by the cylinder and functions:

$$1 = x^2 + y^2, \quad z = 0, \quad z = 1 + x.$$

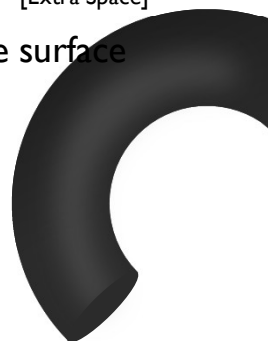


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[Extra Space]

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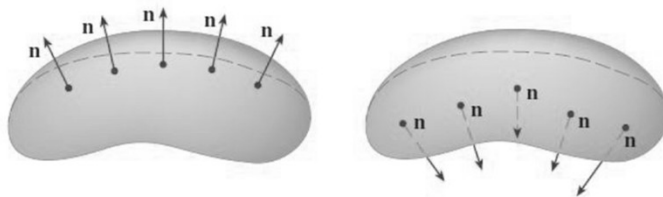


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Orientation

- For surface integrals of a vector field $\vec{F}$ over a surface S , the orientation of the surface must be specified, just like the line integrals. The orientation of a surface is specified via its normal vectors at any point.
- For a closed surface S , we define **the positive orientation** of S to mean that all normal vectors to S point outward (with negative meaning normal vectors point inward instead). If it is impossible to have all normal vectors always point either outward or inward, then the surface S is non-orientable.

FIGURE 7
The two orientations
of an orientable surface



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Orientation



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- To define a surface integral of a vector field, we only want to count the portion of $\vec{F}$ that is normal to the surface S at any point. Hence, the dot product again plays a role in the computation.
- Then the **surface integral** of $\vec{F}$ over the surface S is defined as

$$\begin{aligned}
 \iint_S \vec{F} \cdot d\vec{S} &= \iint_S \vec{F} \cdot \vec{n} \, dS \\
 &= \iint_S \vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} \, dS \\
 &= \iint_D [\vec{F} \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}] |\vec{r}_u \times \vec{r}_v| \, dA \\
 &= \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, dA.
 \end{aligned}$$

This is often called the **flux integral** as it measures the flux of $\vec{F}$ across the surface S .

- Recall and compare it with the line integral of vector fields:

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) \, dt.$$

Orientation

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Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$. (Assume S is positively oriented.)

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[Extra Space]

Example: Find the flux of the vector field $F = (0, y, -z)$ across the surface S given by $y = x^2 + z^2$, $0 \leq y \leq 1$. (Assume S is positively oriented.)

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Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

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[Extra Space]

Example: Let F be the vector field $F = (z^2, x^2, y^2)$, and the surface S be given by $z = x^2 + y^2$, $1 \leq z \leq 9$, $y \geq 0$. Assume S is positively oriented. Set up (but do not evaluate).

$$\iint_S \text{curl}(F) \cdot dS$$

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Questions?

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Thank you

Until next time.

